



Explainable AI Decisions for Human-Autonomy Interactions

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Autonomous systems governed by a variety of adaptive algorithms are making an appearance in safety-critical and time-critical domains, such as automobile traffic and aviation. Until autonomous systems are proven and perceived to be as or more adaptable than humans, and resilient in the face of unanticipated faults and variable conditions, humans will have to remain in ultimate control of decision-making, while supported by machine-based information and advice. Human-machine interaction in many domains has numerous well-known difficulties, including lack or excess of trust, both of which can lead to serious problems, especially when human decision makers are overwhelmed with information. Interactions between humans and autonomous systems – a subset of general human-machine interactions -- will be more problematic still. One complication is that sophisticated machine learning systems produce outputs that may be difficult for a human user to interpret. The development of trust has been a major motivation for the area of explainable AI, or XAI. We offer initial observations about the relation between trust and explicability and propose candidate methods for formalizing these notions in technical domains.

Nomenclature

T	=	Total trust in a system
T_j	=	Justifiable trust
T_m	=	Misplaced trust
D_u	=	Unwarranted distrust
$\{A_t\}$	=	The set of all actions the system takes during a given time
$\{A_a\}$	=	The set of all actions accepted by a human (or another agent) during a given time
$\{A_{ap}\}$	=	The set of all actions that resulted in a positive outcome and were accepted during a given time
$\{A_{af}\}$	=	The set of all actions that resulted in a faulty outcome and were accepted during a given time
$\{A_{rp}\}$	=	The set of rejected actions that had positive outcome

I. Introduction

Autonomous systems governed by a variety of adaptive algorithms, including machine learning, are spreading from well-defined control activities, such as manufacturing, to complex activities with multi-faceted human interactions and decision-making, such as those involved in piloting an aircraft². Until autonomous systems are proven and perceived to be as or more adaptable than humans, and resilient in the face of unanticipated faults and variable conditions, humans will have to remain in ultimate control of decision-making, while supported by machine-based information and advice. Human-machine³ (H-M) interaction in many domains has numerous well-known and well-documented difficulties [1], including lack or excess of trust, both of which can lead to serious problems, especially in time-critical and safety-critical situations where human decision makers quickly become overwhelmed with information. In these situations, humans become either reluctant to take advice from machines or lose situational awareness and basic skills via overreliance on machines.

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² <https://www.technologyreview.com/s/603216/5-big-predictions-for-artificial-intelligence-in-2017/>

³ In this paper, we use the term “human-machine” to stand in for the term “human-autonomy” because the former is used more commonly. However, we mean “interaction between humans and autonomous systems”.

The difficulty is exacerbated by diminishing explicability of the more sophisticated machine learning systems (AI⁴). According to DARPA, "...the current generation of AI systems offers tremendous benefits, but their effectiveness will be limited by the machine's inability to explain its decisions and actions to users. Explainable AI will be essential if users are to understand, appropriately trust, and effectively manage this incoming generation of artificially intelligent partners." [2] The growing importance of explicability is emphasized in the recently adopted European Union General Data Protection Regulation [3] that includes a provision for a "right to explanation" of decisions made on the basis of machine learning algorithms.

To clarify XAI, DARPA juxtaposes statements of current classification algorithms ("this is a cat") with expected XAI statements "This is a cat because it has fur, whiskers, and claws; it has this [visual] feature". On interacting with a decision or a recommendation obtained as a result of an XAI-based machine learning process, the user should be able to declare: "I understand why; I understand why not; I know when you fail; I know when to trust you; I know why you erred." [2]

Literature provides a variety of motivations for explicability and related concepts, such as interpretability and transparency. An informative effort toward developing a comprehensive taxonomy of methods in interpretability research, as well as a critical analysis of the literature concerned with supervised learning can be found in [4]. The author notes that "either: (i) the definition of interpretability is universally agreed upon, but no one has managed to set it in writing, or (ii) the term interpretability is ill-defined, and thus claims regarding interpretability of various models may exhibit a quasi-scientific character" and concludes that the latter appears to be the case.

The existence of a broad variety of machine learning models, approaches and applications that provide context for XAI contribute to the lack of universal agreement as to the concepts of model interpretability. Moreover, explicability and trust, as one of XAI's main motivations, are complicated subjects, fraught with subjectivity, especially when viewed in the context of general intelligence. An extensive overview of various aspects of trust can be found in [5] and references therein.

In this paper, we are interested in examining the hypothesis that explicability or interpretability, in general, can promote *justifiable trust* in technical autonomous systems as they interact with humans. Assuming that XAI does assist in developing justifiable trust in human-machine interactions, it is important to adopt productive formalisms for trust and explicability and the relation between them.

We note that the question of explicability in agent-to-agent interactions is also valid in completely autonomous⁵ systems. Safety-critical software that makes decisions autonomously requires checks by other software. Other software agents must be governed by different algorithms, to provide independent checks; hence the need for understanding explicability and "trust" among fully autonomous agents remains.

In this paper, we offer initial observations about the relation between explicability and trust in H-M systems that produce decision-making outputs. We argue for the importance of context; and advance a number of formalisms for using explicability to promote justifiable trust. The resulting principles can be extended to general heterogeneous multi-agent systems. We will exclude the subject of trust in H-M interactions in social settings and will limit the discussion to cyber-physical-human technical systems. In this discussion, we also exclude problems related to data errors.

II. Trust

Because our overarching target is facilitation of H-M interactions via justifiable trust in machine decisions, we have to define trust to study its dependence on explicability. We consider trust in systems that offer advice or directions, such as a system that would propose a new trajectory to an aircraft pilot, as well as completely autonomous systems where the human serves only to give it authority, i.e., to turn it on, with the system then acting according to its own decisions. An example of the latter is the Automatic Ground Collision Avoidance System (Auto-GCAS)⁶. In this paper, we will make simple points about trust in a completely autonomous system using an example of a robotic surgeon⁷, a recently developed autonomous device designed to suture tissues after surgery.

⁴ We use the term "Artificial Intelligence" or AI loosely, to denote algorithms capable of some degree of adaptive learning and decision-making.

⁵ The notion of autonomy has many competing definitions. For the purposes of this paper, autonomy denotes the ability of a software-based system to function independently, combined with the authority to do so, in some domain.

⁶ <http://aviationweek.com/air-combat-safety/auto-gcas-saves-unconscious-f-16-pilot-declassified-usaf-footage>

⁷ <http://spectrum.ieee.org/the-human-os/robotics/medical-robots/autonomous-robot-surgeon-bests-human-surgeons-in-world-first>

Intuitively, trust is acceptance of an advice or an action, based on sufficient confidence in a positive outcome of that advice or action. This gives rise to several questions:

- What is a positive outcome?
- How much confidence is sufficient?
- How does one arrive at sufficient confidence?

A. Positive Outcomes

One definition of a positive outcome is that a model performs according to its specifications. If a model reflects reality sufficiently closely and performs reliably, we can say that the model is trustworthy. It would seem that trustworthiness in operation would obviate the need for explicability. However, all models are approximations and any proof of a model's reliability rests on the validity in a realistic environment of the assumptions that went into the proof. As the environment changes, the assumptions must be continually re-validated.

All aspects of positive outcomes are difficult to capture because some decisions about the quality of an outcome may be subjective. While humans can excel at decision-making when dealing with a few variables and objectives, the ability deteriorates with growing problem dimension. In fact, optimal decisions in large problems are often counterintuitive to humans. Therefore, human decision makers may not accept a perfectly legitimate decision computed by a machine (or another human) as valid if that decision diverges from the decision maker's experience and intuition [6]. Explanations would be beneficial in getting a human operator used to legitimate but non-intuitive machine decisions.

Another factor in evaluating a decision is the possibly incomplete problem definition in terms of specification of the variables, objectives and constraints. Even assuming a complete definition, formulating an appropriate computational problem amenable to machine learning is difficult, while an inappropriate formulation may lead to unsuccessful outcomes. A user may feel the outcome to be unsuccessful when the outcome is legitimate, subject to the limitations in the decision-making problem formulation. Examples of shortcomings and difficulties in formulating correct problems in AI are discussed in, e.g., [7].

In addition to operational environments, the very recognition of successful and unsuccessful outcomes would enter into training environments for human operators; and explicability would be especially valuable for this purpose.

Despite complexities in identifying successful outcomes in general, technical contexts and, particularly, safety-critical and time-critical contexts, allow us to evaluate an outcome of a machine decision-making process, *a posteriori*, according to the following metrics:

1. Did the decision lead to a safe conclusion? (E.g., did the robotic surgeon produce stitches that were limited to relevant tissues?)
2. Was the decision timely? (E.g., did the system provide the pilot with a flyable trajectory in time to avoid a collision?)
3. Was the actual decision close to the predicted decision? (E.g., did the proposed new trajectory respect the requirement of minimizing the delay in reaching the destination while avoiding a conflict?)
4. Was the decision robust? (E.g., did the new trajectory allow for avoiding secondary conflicts down the line? Did the robotic surgeon provide sutures that remained free of leakage after a long period? Did the suture minimize formation of the scar tissue?)

In both sample contexts, the criteria of success are objective and can be quantified. However, there is a clear context-based difference: the criteria for a successful trajectory generation are physics based and can be easily formalized as computable constraints in a decision-making problem, in principle. While criteria for a successful outcome from a robotic surgeon are clear, the quantification is not as easy; for instance, the outcome is only partially observable in real time, with some of the consequences manifesting only in the future, with non-obvious causality.

B. Sufficient Confidence

The confidence threshold sufficient to declare trust depends on the risk involved in the context of the problem. The risk, in turn, is a function of the uncertainty in the outcome, times the importance of the consequences.

In the example of the robotic surgeon, the device was allowed complete autonomy in 60% of the trials, with the remainder done with input from human physicians. In all cases, the device was given authority to apply sutures to *ex vivo* tissue in a lab and to anesthetized *vivo* tissue, with experienced human surgeons given the same tasks. The robotic surgeon outperformed its human counterparts, producing stitches that were more consistent and more resistant to leaks.

The problem is that, even given the accumulated data of the robotic surgeon's ability to deliver superior stitching, successful outcomes cannot be guaranteed. If the device is approved for clinical use, the confidence in a

positive outcome will be high enough to allow the device to function, but not high enough to allow it to function unobserved. Thus, in a safety-critical context, the threshold of confidence depends on the user of the system and has to be set externally, e.g., via a certification requirement.

The question of public trust, given a certification requirement, is outside the scope of this discussion. However, it is partially addressed here implicitly, because justifiable trust in a technical context goes to trustworthiness, which in turn is one of the components in developing public trust.

C. Arriving at Sufficient Confidence

Achieving confidence in performance requires both analysis and experimentation. Analysis may prove that an algorithm converges under a number of assumptions; experiments must demonstrate the validity of the assumptions in an operational environment. Although much can be done to test algorithms in simulation, realistic validation is unavoidable. Standard statistical results indicate how many experiments must be performed to attain a given degree of confidence; e.g., [8].

Machine decision-making exhibits an additional factor, common to all decision-making approaches, but of particular import in safety-critical applications: crucial dependence on problem formulation. In a recent example [9], the authors discussed a machine learning model trained to predict when a pneumonia patient can be released with the lowest risk of complications. The model concluded that pneumonia patients who also had asthma could be released with the least amount of risk. In reality, the combination of pneumonia and asthma is life threatening. On examination of the result, it turned out that the reason the algorithm connected this high-risk combination with the lowest risk was that such patients received intensive treatment and, thus, tended not to suffer from complications.

The algorithm performed as asked: it identified the patients with the lowest risk of complications. The fact that the outcome was wrong had to do with an error in the problem formulation. In such situations, it appears that explicability in training and operation is a valuable (and sometimes the only) tool for detecting potential problem formulation errors, aside from detecting *a posteriori* effects of an ill-posed problem in operation.

D. Measuring Trust in Experiments

Aside from approaches to achieving trust, one prospective way to measure it is the ratio

$$T = \frac{|\{A_a\}|}{|\{A_t\}|} \quad (Eq. 1)$$

Here $\{A_a\}$ is the set of accepted actions (advice, directions, or operations) and $\{A_t\}$ is the set of all actions that took place during a given time. By this definition, trust is a real number that takes value in the interval $[0, 1]$. The value of zero denotes the lack of trust and the value of one denotes complete trust. Other ranges are possible. One could, for instance, allow negative values to signify various degrees of distrust.

This definition may be particularly simple and useful when we estimate trust thresholds during training autonomous systems via multi-agent simulations, including human-in-the-loop experiments. However, the numerator of Eq. 1 may include actions that led to undesirable outcomes and are therefore undeserving of trust. Let us define justifiable trust as the ratio

$$T_j = \frac{|\{A_{ap}\}|}{|\{A_t\}|} \quad (Eq. 2)$$

In Eq. 2, the set $\{A_{ap}\}$ denotes accepted actions that had a positive outcome. In addition, the following ratio defines misplaced trust:

$$T_m = \frac{|\{A_{af}\}|}{|\{A_t\}|} \quad (Eq. 3)$$

Here, $\{A_{af}\}$ is the set of faulty actions that were accepted. Thus, the total demonstrated trust is

$$T = T_j + T_m. \quad (Eq. 4)$$

We may also consider various metrics for distrust. For instance, unwarranted distrust, as a target of minimization, may be:

$$D_u = \frac{|\{A_{rp}\}|}{|\{A_t\}|} \quad (Eq. 5)$$

Here, the set $\{A_{rp}\}$ contains rejected actions that had positive outcome.

Ultimately, we would like to drive T_m and D_u to zero, so that the total trust value is justified and actions with positive outcome are not rejected unnecessarily. We would also like to express the sets of actions as functions of explicability. This would allow us to evaluate predicted justifiable trust and potentially to design strategies for its maximization, as well as for minimization of unwarranted distrust. A path to quantifying explicability is described in the next section.

III. Explicability

The notion of explicability presupposes that recipients of explanations find their own decisions explicable, and so would find others' explanations convincing if they fit within their own explicability framework. However, in general, human decision makers may not be able to explain how they came up with a specific decision. That is, they may conjecture a model of their own decision-making *a posteriori*, but this model may not describe accurately or at all the real decision-making process that took place. Humans exhibit large variability in the quality of decision-making, especially in the face of too much information. See, for example, [10] for instructive examples of these phenomena. The questions are: what explanations given by algorithms would a human find acceptable? What explanations would be useful to promote trustworthiness of the human-machine interaction?

In the literature, explanations usually fall under one of two categories. One line of inquiry proposes the need for understanding the structure and functioning of the algorithms involved in decision-making. The other disregards the algorithmic structure and looks for explanations that can be derived from the relation between input and output data.

E. Transparency

Machine learning algorithms usually operate as black boxes. Transparency calls for understanding and the ability to explain and possibly replicate the structure of an algorithm at various levels, from the component mechanisms of the training algorithm to the entire model. In essence, transparency implies the ability to reverse engineer an output (e.g., a decision) produced by a learning algorithm. Various aspects of transparency, such as amenity to simulation, decomposability and algorithmic transparency, are described in [2] in detail.

Aside from the potential complexity of assuring such transparency, the notion has two main difficulties. First, humans as decision makers lack algorithmic transparency and thus may lack a reference point for estimating the reasonableness of an algorithm structure. They may not find transparency in machine learning algorithms illuminating and thus conducive to establishing trust. Second, in order to achieve transparency, humans may have to simplify the algorithms, thus potentially reducing their predictive properties and overall utility.

F. A Posteriori Explicability

This approach preserves the black box and attempts to derive explanations from the relation between inputs and outputs and express them as narratives, images, analogies. The DARPA example ("This is a cat because...") is an instance of a posteriori explicability.

While this approach appears more promising than transparency, it has potential difficulties as well. An interesting example is given in [11], where studies in human attention indicate that attention models in Visual Question Answering do not look at the same regions as human, while arriving at the same conclusions. Thus, an explanation of a model may not be convincing to a human if it is not supported by experience. However, in such a case an explanation combined with reliable performance may still promote justifiable trust in a system.

G. Formalizing Explicability

Suppose an algorithm that offers trajectories to pilots proposes a new trajectory. A pilot might find explanations of the following type convincing:

- The new trajectory is offered because the old trajectory is infeasible because
 - There is a violent weather (or another obstacle) in the path of the old trajectory and the new trajectory is flyable and conflict-free
 - The old trajectory will put the aircraft in dangerous proximity to dense traffic; the new trajectory is flyable and conflict-free
 - ...

- The new trajectory is better than the old trajectory because
 - Given local winds, the new trajectory is flyable, conflict-free and will save more fuel
 - A region of airspace with previously dense traffic is now sparse and the new trajectory is flyable, conflict-free and will get the pilot to destination in less time
 - ...

Many other scenarios are possible. Arguably, all acceptable explanations will have a number of common features: the old trajectory is rejected because it violated a number of constraints; the new trajectory is flyable and conflict-free, thus satisfying a set of constraints; some attributes associated with trajectories, expressed as constraints and objectives, have better values than those of the old trajectory.

Regardless of the structure of the algorithm used to compute the trajectory (a nonlinear programming method that computes an optimal trajectory directly or a machine learning approach that proposes trajectories based on data of previously observed trajectories), the proposed trajectory must have attributes that are meaningful to a human decision maker in terms of the physics of the trajectory and value functions associated with moving from point A to point B. E.g., minimizing fuel expenditure, minimizing delays in reaching destination, maintaining safe distance from other aircraft, objects, and weather, and so on.

Thus convincing explanations would contain information about constraint violations and comparative values of the objectives and constraints between the current and proposed decision, such as a trajectory.

Similarly, if a machine learning system is used as a design tool, a proposed artifact must have associated constraints satisfied and objectives at values that are better than those of the previous design.

In the DARPA example of cat classification, the explanation “this is a cat because it has whiskers and fur” would be insufficient. The proposed formalism would require a statement quantifying, e.g., the average distance between the shape cat’s ears and that of a number of other animals.

IV. Optimizing Justifiable Trust

In this section, we outline an approach to computational investigation of trust and its relation to explicability of decisions.

In step one, we formulate a decision-making problem that is meaningful to the user. For instance, in the case of trajectory computation, the problem will be as follows:

$$\min_{x \in \mathcal{C}} \sum_{i=1}^N w_i f_i(x) \quad (Eq. 6)$$

where x is a vector containing trajectory parameterization; the set $\mathcal{C}$ denotes constraints that govern the physics of the aircraft and provide safety buffers; w_i are the weights denoting relative importance of the objective functions to the user; and $f_i(x)$ are the objective functions, such as fuel expenditure and time to destination; plus a set of additional explicability objectives. These objectives contain appropriate norms of the differences between the previous and the new trajectories, the previous and the current values of the arrival delay, and so on.

In step two, we interrogate the black box model to arrive at an estimate of the weights, using interactive multiobjective optimization (e.g., [12]), thus deriving the implicit optimization problem solved by the black box. Inverse reinforcement learning [13] is another promising approach for identification of the objectives.

Now, we build models of trust defined in equations 1-5 as a function of explicability, computed from the objective functions and constraints identified in step 2, starting with a linear model. Simulations serve to provide data for parameter identification of the model. The impact of explicability on the acquisition of trust can then be studied in context.

Finally, given the model that best captures the dependence of justifiable trust of explicability, we can experiment with optimizing the model explicitly, in terms of explicability parameters.

V. Concluding Remarks

In this paper, we discussed a number of initial observations on the hypothesis that interpretability, or explicability, of decision-making algorithms is conducive to the development of justifiable trust in human-machine interactions. While explanations alone may be insufficient to ensure trust, explicability or interpretability has an important role during both training and operation of autonomous systems that interact with human operators. Context is essential in developing actionable metrics of trust and explicability. Our next step is the development of the outlined concepts in the context of mission planning and execution for multi-vehicle trajectory operations [14],

where explicability as an approach to developing justifiable trust will be explored in interfaces between machine learning algorithms and human mission directors.

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